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EX NO: 13	V2X Latency Comparison

AIM: To simulate and compare the performance of a Vehicle-to-Everything (V2X)

communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%),

and Vehicle Speed (km/h) using MATLAB.

Procedure:

1. Define the V2X communication parameters such as vehicle speed, transmission range,

packet size, and simulation time.

2. Generate different vehicle movement scenarios by varying the vehicle speed.

3. Simulate wireless communication between vehicles using a V2X communication model.

4. Calculate the End-to-End Latency (ms) for data transmission between communicating

vehicles.

5. Compute the Packet Delivery Ratio (PDR %) by comparing the number of successfully

received packets with transmitted packets.

6. Analyze the effect of increasing Vehicle Speed (km/h) on communication latency and

packet delivery performance.

7. Plot MATLAB graphs for End-to-End Latency, Packet Delivery Ratio, and Vehicle

Speed.

8. Compare the obtained results to evaluate the efficiency and reliability of the V2X

communication system under different traffic conditions.

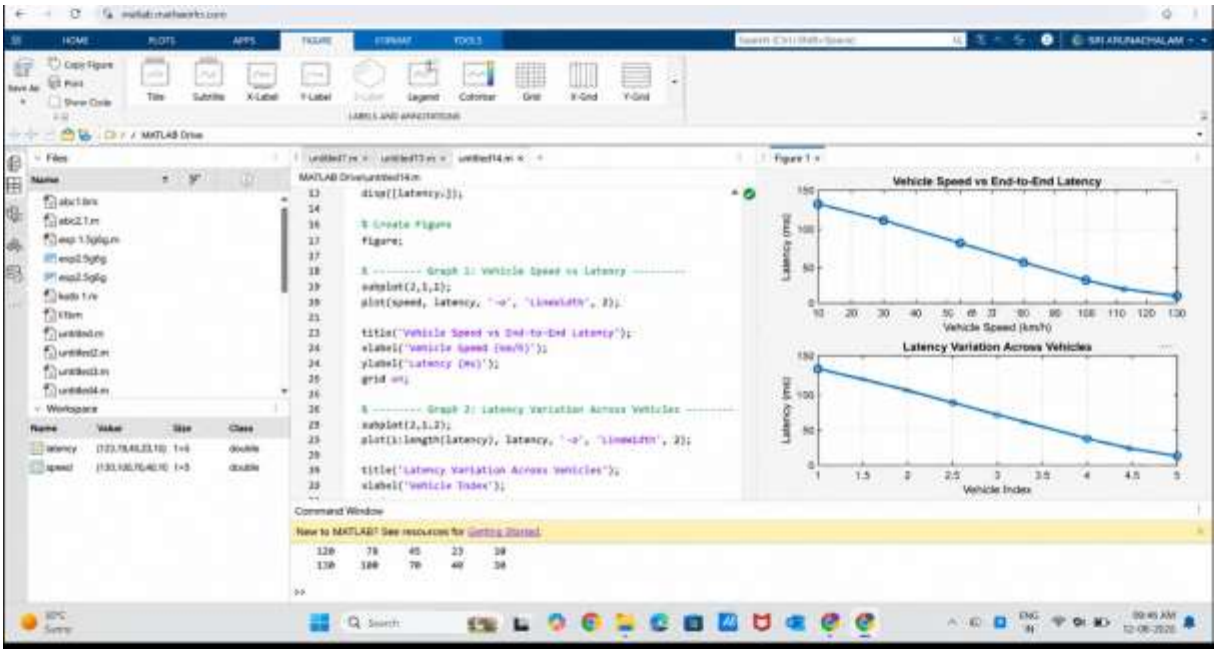
Objective 1 Matlab Code and Output:

```
clc;
```

```
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])

figure
% ------ Graph 1 ------
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% ------ Graph 2 ------
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```

OUTPUT:



Result:

The End-to-End Latency decreases as the vehicle speed increases under the simulated communication conditions. Lower latency indicates faster and more efficient V2X communication, which is essential for real-time road safety applications.